AN INTEGRATED SMART SYSTEM FOR CASHEW CULTIVATION

Project ID - 25-26J-175

Project Proposal Report

Sanoch M – IT22159694

B.Sc. (Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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DEVELOPMENT OF AUTOMATED GRADING SYSTEM FOR CASHEW NUTS.

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Date

ABSTRACT

This paper presents the development of an Automated Cashew Nut Grading System, a crucial and innovative component of a comprehensive Integrated Smart System for Cashew Cultivation. The system is specially designed to propose the significant challenges and inefficiencies associated with traditional manual cashew grading, which is a labour oriented, time consuming, and subjective process leads to human error.

Our approach centres on a computer vision based model that enables a farmer to simply take a photo of the harvested nuts and feed it into the system for instant analysis. The system, trained on a variety of key external features, including its size, shape, colour, and texture defects. This allows for a precise and objective classification and grading process, where nuts that do not meet the quality standards are identified and can be subsequently rejected. This automated method not only reduces the need for extensive manual labour but also ensures consistent quality, which is vital for maintaining a competitive edge in the global market.

Beyond its primary function as a post-harvest tool, this system is designed to provide a critical feedback for the entire cultivation lifecycle. The detailed quality data collected from the grading process is not isolated, it is stored in a database, where it can be used for future analysis along with other sub system, such as soil testing, disease detection, pest identification. This approach empowers farmers to transition from a reactive model of problem solving to a proactive and data driven one. By combining the final product quality with specific environmental and cultivation conditions, farmers can gain valuable insights to optimize their farm management practices and leads to improved crop yield, reduces waste, and enhanced profitability. This innovative framework transforms the act of grading into a powerful tool for continuous agricultural improvement.

Keywords: Automated Cashew Nut Grading System, post-harvest tool, soil testing, disease detection, pest identification

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List of Abbreviations

Abbreviation	Description
CNN	Convolutional Neural Networks
AI	Artificial Intelligent
DL	Deep Learning
ML	Machine Learning
KNN	K-Nearest Neighbors

1. INTRODUCTION

Cashew nut grading plays a critical role in determining market value and ensuring consistent quality for export and domestic consumption. Traditionally, grading has been conducted manually, relying on visual inspection by trained workers. While effective in certain contexts, this approach is labour intensive, subjective, and prone to inconsistency, leading to variations in quality assessment and fluctuations in farmer income. Research in agricultural post-harvest automation has shown that computer vision and machine learning can significantly improve the accuracy, repeatability, and speed of grading processes.

The current state of the art includes deep convolutional neural networks (CNNs) [6] and transfer learning models that can classify kernels with over 90% accuracy, and segmentation frameworks capable of detecting surface defects under controlled lighting. Some systems have also explored embedded deployment on low cost devices like Raspberry Pi for real time sorting. However, most existing solutions focus either on visual grading or defect detection, without integrating density estimation or providing a unified grading outcome.

In previous research attempts have addressed the problem by using handcrafted features with classical classifiers or deep CNNs [1] trained on curated datasets. While these approaches improved classification accuracy, they often required separate physical sensors for weight measurement or lacked scalability for farmer use. The proposed approach builds on these foundations by integrating defect detection, multi attribute grading, and density estimation into a single vision-based system, eliminating the need for additional sensors. This integration will offer a portable, low cost, and standardised grading solution with a user-friendly interface implemented for farmers.

1.1 Background

The cashew is a globally significant tree nut, playing a pivotal role in the agricultural and economic landscapes of many developing nations. Renowned for its unique flavour, nutritional value, and versatility, the cashew kernel is a highly sought after commodity in international markets. “An Integrated Smart System for Cashew Cultivation”, a critical bottleneck in the cashew value chain is the post harvest processing, particularly the grading of cashew kernels. This project, which focuses specifically on the “Development of an Automated Grading System for Cashew Nuts,” addresses the significant challenges associated with traditional manual grading a process that is not only labour oriented and time consuming but also prone to human error and inconsistency [1,2].

The economic viability and competitiveness of the cashew industry depend directly on the quality of the final product. Cashew kernels are classified into different grades based on a variety of attributes, including size, colour, and the presence of defects. Whole kernels command a premium price, while broken or discoloured nuts are sold at a significant discount [7]. This fundamental relationship between quality and market value underscores the urgent need for a precise and efficient grading system. The existing manual process, however, is a major impediment to achieving this goal. This background section will explore the problems inherent in manual cashew grading, the economic importance of quality, and the strategic rationale for developing an automated, computer vision-based solution.

The traditional method of cashew grading relies heavily on the subjective judgment and physical labour of workers. This manual process involves painstakingly sorting each individual nut by hand, a task that is fraught with challenges and inefficiencies.

The most prominent issue with manual grading is its lack of standardization. Quality assessment is based on a worker’s visual perception of a nut's physical characteristics, such as shape, size, colour, and the presence of defects. The subjective nature of this visual inspection means that different workers may apply different criteria, leading to a lack of uniformity in the final product [9]. Defects like brown patches, scraped surfaces, or discoloration often go unnoticed or are misclassified, directly impacting the final grade and marketability of the product.

Manual grading is inherently slow and cannot keep pace with the high volume of production required to meet global demand. This limits the throughput of processing plants and creates bottlenecks in the supply chain. The need for a large workforce to perform this tedious task also leads to substantial labour costs, which can erode profit margins for processors, particularly small and medium-sized enterprises [10]. In an industry where profitability is tied to efficiency, the slow speed and high cost of manual labour are unsustainable. The project aims to overcome this by introducing a system that can process a large volume of nuts rapidly and accurately, thereby boosting productivity and reducing operational costs.

Beyond the economic and efficiency concerns, manual cashew processing poses serious health risks to workers. The cashew nutshell liquid (CNSL), a corrosive and phenolic resin found between the shell and the kernel, is highly toxic and can cause severe skin irritation, burns, and other dermatological issues [11]. Workers who manually cut and handle the nuts are at a high risk of developing contact dermatitis, skin ulcerations, and other complications from prolonged exposure. Furthermore, the repetitive motions and awkward postures required for manual sorting can lead to musculoskeletal disorders. The development of an automated system will significantly reduce the human-to-nut contact, thereby enhancing worker safety and improving the overall working environment [10].

The global cashew market operates on a rigorous system of grading and standards that directly link quality to price. Buyers in key markets like the U.S., Europe, and the Middle East have specific preferences for cashew grades, and adherence to these standards is essential for successful trade [7]. Kernels are categorized as "whole" or "broken." Whole kernels, such as the premium W180, W210, and W320 grades (referring to the number of kernels per pound), fetch the highest prices. Broken grades, including butts, splits, and pieces, are sold at a lower value [7, 8]. The goal of a processing plant is to maximize the yield of whole kernels. Manual processing often results in a higher percentage of broken nuts, reducing overall profitability.

The colour of the kernel is another crucial indicator of quality. Cashews are graded based on their colour, with "white" or "pale ivory" kernels considered superior. Kernels with a yellowish tint or a "scorched" appearance are of a lower grade and value. The "dessert" grade, which includes deeply scorched or discoloured kernels, is the lowest grade [8]. The presence of defects, such as blemishes, spots, or damage, also affects a nut's grade. International standards, such as those published by the UNECE, specify permissible defect tolerances for each class of kernel. These defects can arise from improper handling, overheating, or contamination, and an effective grading system must be able to accurately identify them [9].

1.2 Literature Survey

Cashew nut processing after harvest is crucial, yet it faces challenges with outdated manual grading techniques. These traditional methods are not only time consuming but also rely heavily on human judgement, which can be inconsistent and error prone. As a result, the quality of cashew nuts varies, lowering their market value and increasing waste. The lack of precision in manual sorting has fueled the need for smarter, automated solutions. To address this, researchers are turning to advanced technologies like computer vision and AI. These innovations promise faster, more accurate grading while reducing human effort. The shift toward automation could revolutionize the cashew industry, ensuring better quality and higher profits.

Early research efforts focused on developing cost effective and basic machine learning solutions. Let's breakdown some research papers related to my component.

Cashew nut grading has shifted from manual inspection to automated machine learning and deep learning techniques. Earlier studies used models such as SVMs, Naïve Bayes, and neural networks, mainly relying on colour, size, shape, and texture features. These methods [4] achieved moderate accuracy levels ranging from 76%–86%, with later improvements through CNN-based models like CashNet-15 that classified nuts into categories such as intact, whole, split, and broken. While these approaches showed promise, they were often limited in handling broken or low-quality samples and lacked real-time deployment in actual grading systems. The latest advancement is a low-cost automated system powered by a YOLOv8–Transformer hybrid model, where nuts are captured on a conveyor by a camera and classified into four grades: good, error 1, error 2, and error 3. The system integrates spatial pyramid pooling and Transformer blocks for robust feature extraction, while pneumatic actuators automatically sort the graded nuts. Trained on 6000 images, the model achieved 96% precision and recall with a mean average precision of 98.4%, outperforming other YOLOv8 variants. Testing on unseen samples further confirmed its strength, with only a 2.96% error rate, making it a state-of-the-art and cost-effective solution for cashew nut grading

The grading of pistachio seeds is carried out by examining characteristics such as size, shape, colour, and overall quality, but manual sorting is inefficient and inconsistent, creating a need for automated solutions [5]. Early works applied CNN-based architectures including Alex Net, VGG16, and VGG19, where Alex Net underperformed with 94.42% accuracy while VGG16 and VGG19 achieved stronger results. InceptionV3, however, produced only 61.75% accuracy, showing its limitations for this task. Other efforts such as EfficientNetB1 successfully classified four pistachio types (long, jumbo, super long, and round) with high accuracy, while Retina Net reached 94.75% accuracy across six categories. Similarly, ResNet models delivered 88.57% accuracy, and hybrid methods like autoencoders and ANN with random forest showed accuracies of 80.3% and 93.65%, respectively. More recent work emphasizes lightweight and efficient models for practical deployment. A fine-tuned MobileNetV2 model has been proposed as a state-of-the-art solution, making use of bottleneck layers and shortcut connections to reduce computational cost while

preventing vanishing gradients. Tested on the Kaggle pistachio dataset with Siirt and Kirmizi classes, the model achieved 95.81% training accuracy and 95.57% testing accuracy, outperforming all earlier approaches. This demonstrates that MobileNetV2 provides both efficiency and high precision, making it highly suitable for real-time pistachio grading applications

Cashew nut grading has long been done manually by laborers who sort based on visual attributes such as shape, size, and colour, but this approach [1] is inefficient, error-prone, and inconsistent. To address these limitations, automated grading systems using computer vision and machine learning have been developed. The grading workflow typically includes four phases: image acquisition with cameras, pre-processing to remove noise and enhance quality, segmentation to separate the nut from the background, and classification to determine whether the nut is whole, half, or defective. Features such as HSV-based colour moments, texture (energy, contrast, correlation, homogeneity, entropy), and geometric properties are extracted and used for classification. Traditional models such as SVM and artificial neural networks have been applied, though they rely heavily on handcrafted features and achieve only moderate accuracy. Recent approaches leverage deep neural networks and transfer learning, which automatically extract hierarchical features and yield higher precision. In one study, a deep CNN-based grading system implemented in MATLAB achieved 100% training and validation accuracy and over 80% testing accuracy, highlighting its advantage over manual grading. The model was integrated with a Raspberry Pi-controlled conveyor and sorting mechanism, allowing for real-time classification and mechanical segregation of cashew nuts based on quality. This demonstrates that computer vision combined with deep learning not only improves grading accuracy but also provides a scalable and cost-effective solution for the cashew processing industry

Cashew grading is carried out by capturing images of kernels and analysing features such as colour, size, shape, and texture to classify them into grades like whole, split, scorched, or defective[2]. The process generally involves image acquisition, pre-processing to remove noise, feature extraction, and classification, followed by automated sorting mechanisms. Earlier approaches relied on statistical and machine learning models such as Random Forest, Naïve Bayes, K-Nearest Neighbour, Decision Tree, and Multilayer Perceptron, which achieved moderate accuracies in the range of 76–86%. Later studies integrated Support Vector Machines (SVM) and Backpropagation Neural Networks (BPNN) using features like colour, shape, and texture, reaching accuracies of 97.7–99.6%, while regression-based models achieved around 83.5%. With the advancement of deep learning, CNN-based models such as Inception-V3, ResNet50, VGG-16, and optimized frameworks like CashNet-15 have been applied, achieving improved classification performance. For instance, CashNet-15 with 15 convolutional layers reached 97.7% accuracy, while Inception-V3 and ResNet50 reported around 95.1% accuracy. Other models like ANN and fuzzy logic-based classifiers showed accuracies between 86–94%, depending on the features used. Overall, the literature shows that grading accuracy has improved significantly from early ML approaches to deep CNN-based methods, with the best-performing systems

reaching over 97% accuracy, though real-time detection and handling of damaged or split kernels remain ongoing challenges.

Cashew grading is carried out by capturing images of kernels and analysing features such as size, shape, colour, and texture to classify them into categories like whole, broken, split-up, split-down, and defect [3]. Traditionally, manual grading was labour-intensive and inconsistent, leading to the adoption of automated machine vision systems. These systems generally follow four steps: image acquisition, pre-processing, feature extraction, and classification. Earlier methods used feature-based machine learning models such as Random Forest, Decision Trees, Support Vector Machines (SVM), and Backpropagation Neural Networks (BPNN), achieving accuracies in the range of 86–99%, with Random Forest reaching up to 99.8% in some cases. Recent advancements introduced deep learning methods that automatically extract hierarchical features and perform end-to-end classification. Convolutional Neural Networks (CNN) and object detection models like YOLO have been widely applied. In particular, YOLOv5 demonstrated superior performance with 97.65% accuracy and very fast inference (0.025s per image), making it suitable for real-time cashew grading, while CNN models achieved around 97.62% accuracy. YOLOv9 was also tested but underperformed compared to YOLOv5, with an accuracy of 82%. Overall, the literature indicates that the grading process has evolved from manual sorting and traditional ML approaches to advanced CNN and YOLO-based systems, with state-of-the-art models consistently achieving above 97% accuracy, ensuring reliable and efficient classification for industrial applications.

Cashew nut grading is typically performed by capturing kernel images and analyzing attributes such as size, shape, texture, and colour to classify them into categories like whole and split kernels [6]. Traditionally, manual inspection was used, but it is time-consuming, subjective, and prone to inconsistencies. Automated grading systems adopt image acquisition under controlled lighting, followed by pre-processing steps like background removal, grayscale conversion, and feature extraction to prepare the images for classification. Early works applied traditional machine learning models such as Support Vector Machine (SVM), K-Nearest Neighbour (KNN), and Random Forest (RF), which showed reliable performance. For instance, in one study SVM achieved 93% accuracy, outperforming KNN at 91% and RF at 90%, demonstrating its robustness in handling high-dimensional image data and binary classification problems. In recent years, deep learning has emerged as a more powerful approach for cashew grading. Convolutional Neural Networks (CNNs) have been used to automatically learn discriminative features from cashew images, eliminating the need for handcrafted feature extraction. Transfer learning with pre-trained CNNs has also been explored, yielding state-of-the-art accuracy levels that surpass traditional methods. Models such as CNN and YOLO variants have shown accuracy exceeding 97% in classifying cashews into multiple categories including whole, broken, and defective. Overall, the literature highlights that cashew grading has evolved from manual sorting to machine vision–based automation, with SVM and other classical models offering strong baselines, while CNNs and deep learning methods now set the benchmark for achieving above 95% accuracy in real-world grading applications. The literature provides a solid foundation for automated nut grading, but a key research

gap remains in developing a single, integrated system that can non-destructively grade cashews based on a comprehensive set of parameters like size, colour, texture, defects, and inferred density and weight with in a unified computer vision and regression framework.

1.2 Research Gap

Early research heavily relied on models like Support Vector Machines (SVM), Random Forest, and Backpropagation Neural Networks (BPNN) [2, 6]. These models demonstrated reliable performance, with some studies reporting accuracies as high as 99.8% with Random Forest [3]. However, their core limitation lies in the need for extensive, manual feature engineering, requiring researchers to handcraft features such as colour, texture, and shape. This process is time-consuming, subjective, and makes the system less adaptable to new types of defects or variations in lighting.

The field advanced with the adoption of deep CNN architectures like VGG16, VGG19, and ResNet [4, 5]. These models addressed the feature engineering problem by automatically learning discriminative features from the data. While they showed promising results and consistently achieved accuracies above 95% [2, 4], their computational requirements were often high, and some, like InceptionV3, were shown to underperform on specific tasks with an accuracy of only 61.75% on pistachio grading [5]. This indicated that their general-purpose nature wasn't always a perfect fit for a specialized task without significant fine-tuning.

The introduction of models from the YOLO (You Only Look Once) family marked a significant leap forward in both speed and accuracy. YOLOv5, for instance, achieved an impressive accuracy of 97.65% with a very fast inference time of 0.025 seconds per image, making it highly suitable for real-time applications [3]. A hybrid YOLOv8-Transformer model was also noted for its high precision and recall of 96% and a mean average precision of 98.4% [4]. These models are state-of-the-art for fast and accurate classification, but they are primarily designed for object detection, providing only a bounding box around the object. This is a critical functional gap because while they can tell you a nut is "defective," they cannot precisely measure its dimensions.

YOLOv8n-seg, is the superior choice because it overcomes the limitations of all these previous approaches by uniquely integrating the core strengths of each. It provides the automation of feature learning like CNNs, the high speed and accuracy of the YOLO family, and most importantly, it adds the capability of instance segmentation. This allows your system to generate a precise, pixel-level mask of each individual nut. This mask is the foundation for a unified framework that not only accurately classifies nuts but also enables the non-destructive, precise measurement of their length, width, and inferred density, a crucial feature missing from all other models you reviewed.

1.3 Research Problem

One of the main challenges in addressing the research gaps in automated cashew nut grading is the need for a comprehensive system that integrates multiple grading factors such as size, colour, texture defects, and density. Ensuring that all relevant features are effectively combined into one grading model can be difficult, as current systems often focus on limited aspects and may overlook other important features. Additionally, the quality of the dataset used for training the model plays a crucial role in achieving high accuracy, and finding or creating diverse datasets that cover the various varieties and conditions of cashew nuts can be challenging. Another issue is ensuring that the automated system provides accurate grading on par with manual processes, especially for smaller-scale farmers with limited technological infrastructure. Balancing the precision of the system with the affordability and accessibility for farmers is also a key challenge. Lastly, integrating advanced machine learning techniques and ensuring the system is scalable while being cost-effective can be a complex task, as it requires the system to handle a large number of images and features efficiently.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this project is to design, develop, and deploy a comprehensive smart system for cashew cultivation, a transformative solution intended to modernize, strengthen quality assurance, and boost the overall sustainability of Sri Lanka's cashew industry. The system integrates advanced computer vision for both on-the-spot leaf disease and pest identification and a post-harvest grading tool that classifies nuts by defect while simultaneously predicting key physical attributes for precise grading. This is complemented by a soil analysis component to measure nitrogen levels and provide tailored recommendations. This multifaceted platform will provide farmers with data-driven insights and actionable recommendations, with the ultimate goal of improving decision-making, reducing manual labour, ensuring yield consistency, and enabling fair market valuations for their harvest.

2.2 Specific Objective

In this research project, a dedicated component will be developed to address the automation of cashew nut grading. This component focuses on creating an intelligent vision-based system capable of performing a comprehensive, multi-factor analysis of harvested nuts. It aims to eliminate the subjectivity and labour intensive nature of traditional grading by leveraging advanced machine learning to precisely classify nuts based on defects and predict their physical dimensions. This will allow for the automated assignment of specific quality grades, ensuring consistency and a standardized quality control process. The component's specific goals are as follows:

1. High-Quality Dataset Creation and Annotation: A comprehensive and diverse dataset will be collected, consisting of images of cashew nuts from various angles and lighting conditions. This dataset will be meticulously annotated using instance segmentation to create precise, pixel-level masks for each nut. The annotations will also include labels for different defect types (e.g., blemishes, cracks, burns) to ensure the model can accurately identify and classify imperfections.
2. YOLOv8n-seg Model Training and Optimization: The YOLOv8n-seg model will be trained on the annotated dataset. This will involve fine-tuning its architecture and hyperparameters to achieve high accuracy in both object detection and instance segmentation. The model will be specifically optimized to perform efficiently on a resource-constrained edge device, such as a Raspberry Pi, ensuring its viability for real-time, on-site use.
3. Physical Measurement and Grading Framework Development: A crucial part of this component will be the development of a framework to accurately predict

physical attributes. Using the precise pixel masks generated by the YOLOv8n-seg model, a separate algorithm will be developed to non-destructively measure the length and width of each nut. Based on these measurements and the defect classification, a grading system will be implemented to automatically assign a final grade (e.g., Grade A, B, C) to each nut, mirroring industry standards.

4. Integration and Validation: The cashew nut grading component will be integrated into the main smart system. Extensive validation will be performed to assess the system's accuracy and performance in a real-world setting. This will involve comparing the automated grading results with manual expert grading to confirm the system's reliability and to provide actionable feedback for further refinement.
5. User Interface Development: A simple and intuitive user interface will be designed and built. This interface will allow farmers to easily upload images of their cashews and receive an immediate and accurate grade, empowering them with a tool to improve their quality control.

3. METHODOLOGY

The methodology for developing the automated cashew grading system is a systematic process designed to ensure the project's success from initial conception to final deployment. This involves a comprehensive approach, starting with the careful definition of the project's scope, requirements, and target audience. Following this, the system's architecture is designed, outlining the flow of data from image acquisition to the final grade assignment. This includes the selection of appropriate software and hardware technologies, such as a camera for image capture, and deep learning frameworks like TensorFlow or PyTorch for the core processing. The implementation phase focuses on building the system's features, including the creation of a large, annotated dataset of cashew nut images, training a deep learning model for defect detection and dimension extraction, and developing a regression model to predict weight and density. The system then undergoes rigorous testing and validation to ensure its accuracy and reliability, with measurements being compared against manual scales. Finally, the system is deployed and evaluated for its performance in a real-world environment, with ongoing maintenance and updates planned to address bugs and adapt to evolving industry standards, ensuring its long-term relevance and effectiveness.

3.1 System Architecture Diagram

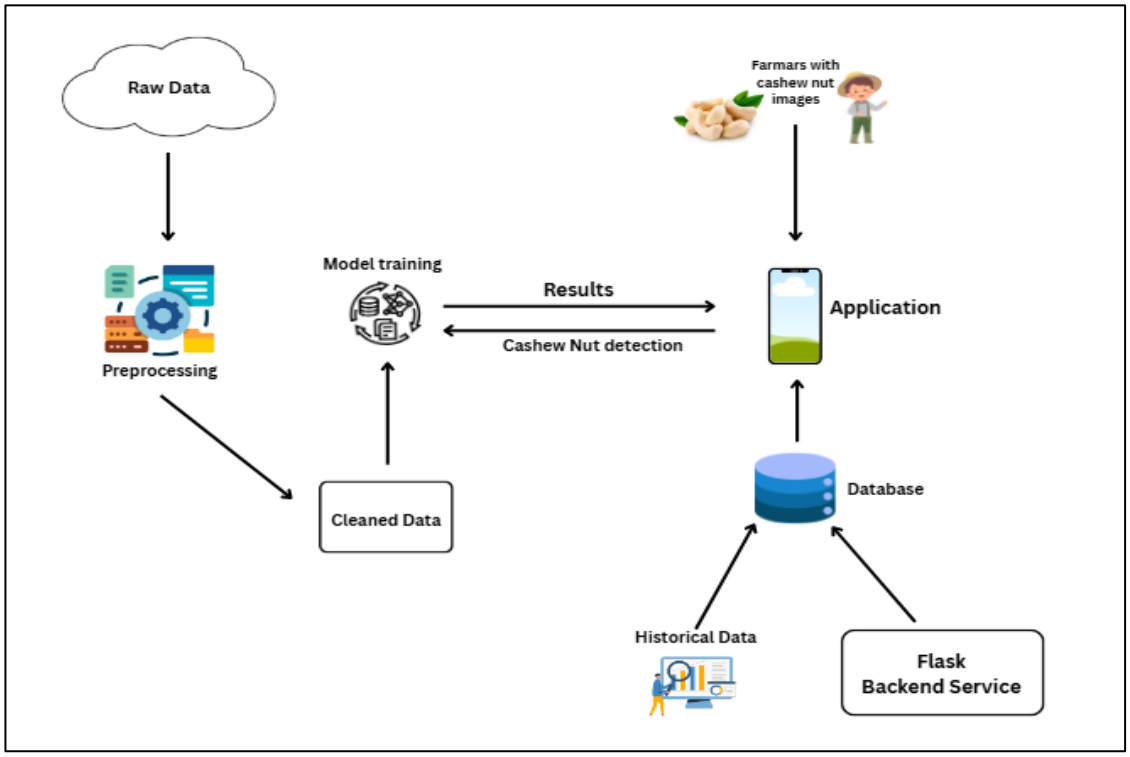

Figure 1.1 System Architecture Diagram

The system's process shown in the Figure 1.1 begins with the Data Acquisition phase, where images of cashew nuts are captured directly in the field using a mobile device. These images are then sent to a central processing unit, which acts as the core of the system. The central unit performs automated analysis on the images, utilizing a pre-trained YOLOv8n-seg model to detect and identify individual nuts. The model doesn't just classify; it also generates a precise, pixel-level mask for each nut, a process known as instance segmentation. This detailed output from the model is then used by subsequent modules to perform their specific tasks. A Grading Framework analyzes the segmentation masks and associated data to classify nuts as defective or undefective and predict physical attributes such as length, width, and density. This comprehensive analysis leads to a final quality grade for each nut. The system's final output—the grading results and any relevant visualizations—is then sent back to the mobile application for the user to view. This entire process is designed to be efficient, providing a seamless experience for farmers. The foundation of your system is the creation of a high-quality, in-field dataset. Images of cashew nuts will be captured under real-world conditions to account for variability in lighting, angle, and clumping. This ensures the model is robust and can handle practical challenges. The images will be meticulously annotated with pixel-level masks for instance segmentation, which is a crucial step for training your YOLOv8n-seg model to be able to identify each individual nut. The YOLOv8n-seg model will be trained on the custom-annotated dataset. You will use data augmentation techniques, such as rotation and brightness adjustments, to enhance the model's ability to generalize and perform well in different environments. This training approach is designed to make the model robust against variations in field conditions, including different lighting and occlusions from other nuts. The model’s efficiency will also be a key consideration, as it needs to be performant enough for real-time use. This is the unique aspect of your component. Instead of just providing a bounding box, the model’s instance segmentation masks will be used to accurately calculate the physical dimensions of each nut, such as its length and width. This physical data, combined with defect classification, will feed into a comprehensive grading framework that automatically assigns a grade (e.g., Grade A, B, C), providing a standardized and objective quality assessment.

3.2 Component Architecture

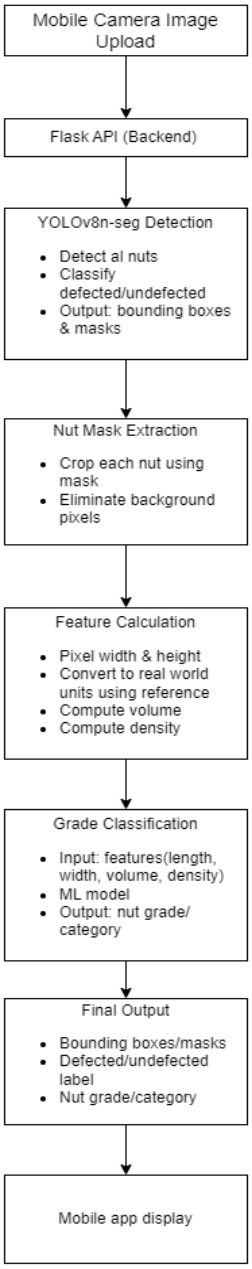

Figure 1.2 Component Architecture Diagram

The architecture in above Figure 1.2 begins with an input image captured via the mobile app, which is processed by the YOLOv8n-seg model to detect nuts, classify them as defected or undefected, and generate precise masks. Each nut is then cropped using the mask, eliminating background pixels. From these segmented nuts, features such as length, width, volume, and density are calculated and fed into a classification model that assigns grades or categories. Finally, the results, including defect status and grade, are displayed on the mobile app and can optionally be stored in a database.

3.3 Functional Requirement

1. Lighting level
   - Neutral white with high colour fidelity for reliable colour features. The illumination at the nut surface should be maintained in the range of 1000–2000 lux, providing sufficient brightness for defected nuts without introducing noise.

2. Image resolution
   - The resolution must be high enough to preserve fine details such as cracks, peeling, and texture variations that are essential for accurate grading. A minimum of megapixels is recommended.

3. Offline Predictions
   - The app must perform the nut grading and display the results without requiring an internet connection.

4. Image Capture & Storage
   - Enable direct image capture from mobile cameras with local storage; supports optional server synchronization when online.

5. Previous/historical data access
   - The system must be enabling the user to get decisions by reviewing the previous grading results.

6. Phone specification
   - The recommended phone operating system is Android.

3.4 Non-Functional Requirements

1. Accuracy
   - The system's grading accuracy must be at least 95%, comparable to expert manual grading.
2. Performance
   - The system should process and grade a single image within a few seconds.
3. Usability
   - The user interface should be intuitive and easy for a farmer with minimal technical expertise to use.
4. Cost effectiveness
   - The hardware and software components should be affordable, particularly for small scale industries.

5. Scalability
    - The system should be able to handle a large number of images and be adaptable to new cashew varieties.

3.5 Expected Test Cases

1. Test Case 1
   An image of a perfectly graded cashew nut is uploaded. The system should correctly identify it as defect free and assign the highest grade.

2. Test Case 2
   An image of a perfectly graded cashew nut is uploaded under proper lighting conditions (1000–2000 lux). The system should correctly identify it as defect-free and assign the highest grade.

3. Test Case 3
   An image of a cashew nut with a visible defect is uploaded. The system should correctly identify the defect and assign a lower grade accordingly.

4. Test Case 4
   An image of a small cashew nut is uploaded. The system should accurately extract its dimensions and predict a corresponding low weight and small size grade.

5. Test Case 5
   A user uploads a batch of images and the system processes them successfully, displaying the grades for each nut in the user interface.

3.6 Technology Stack

1. Programming Language
    - Python: Used for the development of machine learning models, image processing, and the backend logic.

2. Machine Learning and Deep Learning
    - YOLOv8/YOLOv8n- seg (segmentation) model
    - Google Colab- Training and evaluating ML models.

3. Computer Vision
    - OpenCV: A widely used computer vision library for image processing tasks such as contrast enhancement, segmentation, feature extraction, and colour analysis.

4. Backend Frameworks
    - Flask: A lightweight python web framework used to develop the backend of the system, enabling farmers to upload images and receive grading results through a web based interface.

5. Frontend Frameworks
    - HTML/CSS/JavaScript
    - React Native/Flutter

6. Designing Tools
    - UI design: Figma
    - Diagram designs: Draw.io, Canva

7. Database
    - SQLite/MySQL/MongoDB

8. Deployment and Hosting
    - AWS/Google Cloud/Azure

9. Version Control
    - Git/GitHub

3.7 Gantt Chart

The following Gantt chart shown in the Figure 1.3 details a 12-month software research project timeline. It begins with a Feasibility Study (Jun-Jul) for background research, requirement gathering, and proposal approval. The Design phase (Aug) creates the system architecture. Implementation (Sep-Nov) covers development and integration, followed by rigorous Testing (Dec-Jan). The final stages involve Deployment, Maintenance (Feb-Apr), and project Presentations, concluding with Final Documentation and Submission in May. The chart shows a logical sequence from conception to delivery, with key phases like testing and documentation overlapping development.

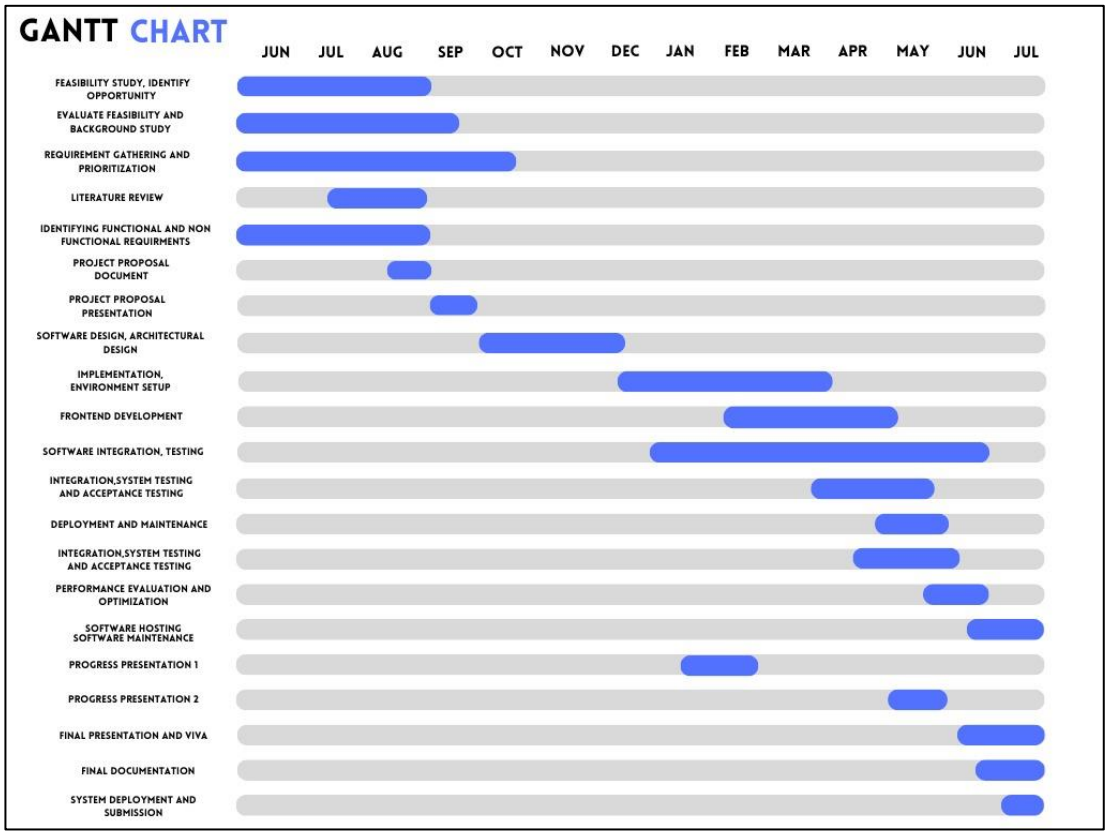

Figure 1.3 Gantt Chart

3.8 Work Breakdown Structure

Based on the provided structure shown in the Figure 1.4, this Work Breakdown Structure (WBS) outlines the hierarchical decomposition of the Cashew Grading System project into manageable phases and deliverables. It begins with the Planning phase, which encompasses initial definition activities like scoping, timeline creation, and charter development to establish project foundation. This is followed by the Analysis stage, focused on gathering requirements, understanding the existing system, and identifying errors to be resolved. The Design phase then details the architectural, interface, and database plans, translating requirements into technical blueprints. Subsequently, the Implementation stage covers the core data science workflow: collecting a dataset, creating, training, and testing the machine learning model for grading. Finally, the Launch phase involves deploying the application, promoting it, and releasing the final product to market, ensuring a transition from development to operational use.

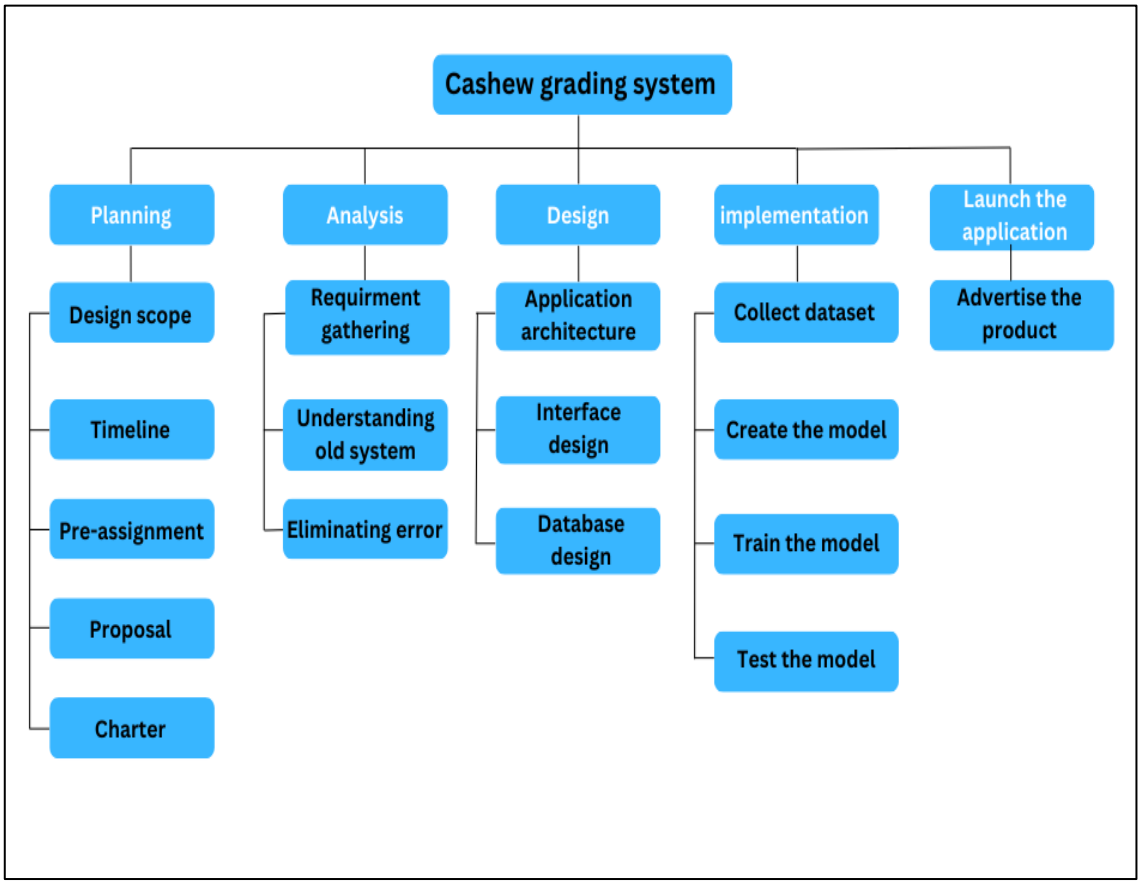

Figure 1.4 Work Breakdown Structure

4. CONCLUSION

Our Automated Cashew Nut Grading System is a crucial component that transforms the traditional, labour-intensive post-harvest process into a precise, data-driven operation. The core of this system is simple: a farmer captures an image of the cashew nuts, and our component's trained computer vision model instantly analyses the image to classify and grade the nuts. This provides an objective and efficient method for quality control, replacing subjective manual sorting.

However, the true value of this component extends beyond simple classification. As part of the Integrated Smart System for Cashew Cultivation, the grading system serves as a vital feedback mechanism. The data on nut quality, defects, and rejections is not isolated; it is integrated into the central data hub. This allows the system to correlate post-harvest outcomes with cultivation-phase data from other components, such as pest detection or soil monitoring. For instance, if the system consistently identifies a high rate of low-grade nuts, it could be a direct indicator of a specific issue during the growth cycle.

In essence, our component empowers the farmer to move from merely reacting to problems after the harvest to proactively optimizing the entire cultivation process. By providing a clear and precise final assessment of the crop, it closes the loop of the smart system, allowing for continuous improvement in yield quality and farm profitability.

5. BUDGET

Item	**Description**	**Cost Estimation**
Deploy in Play store	Deploy the app in the play store	7550 LKR
Google Cloud	12-months	9100 LKR
Total		16,650 LKR

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